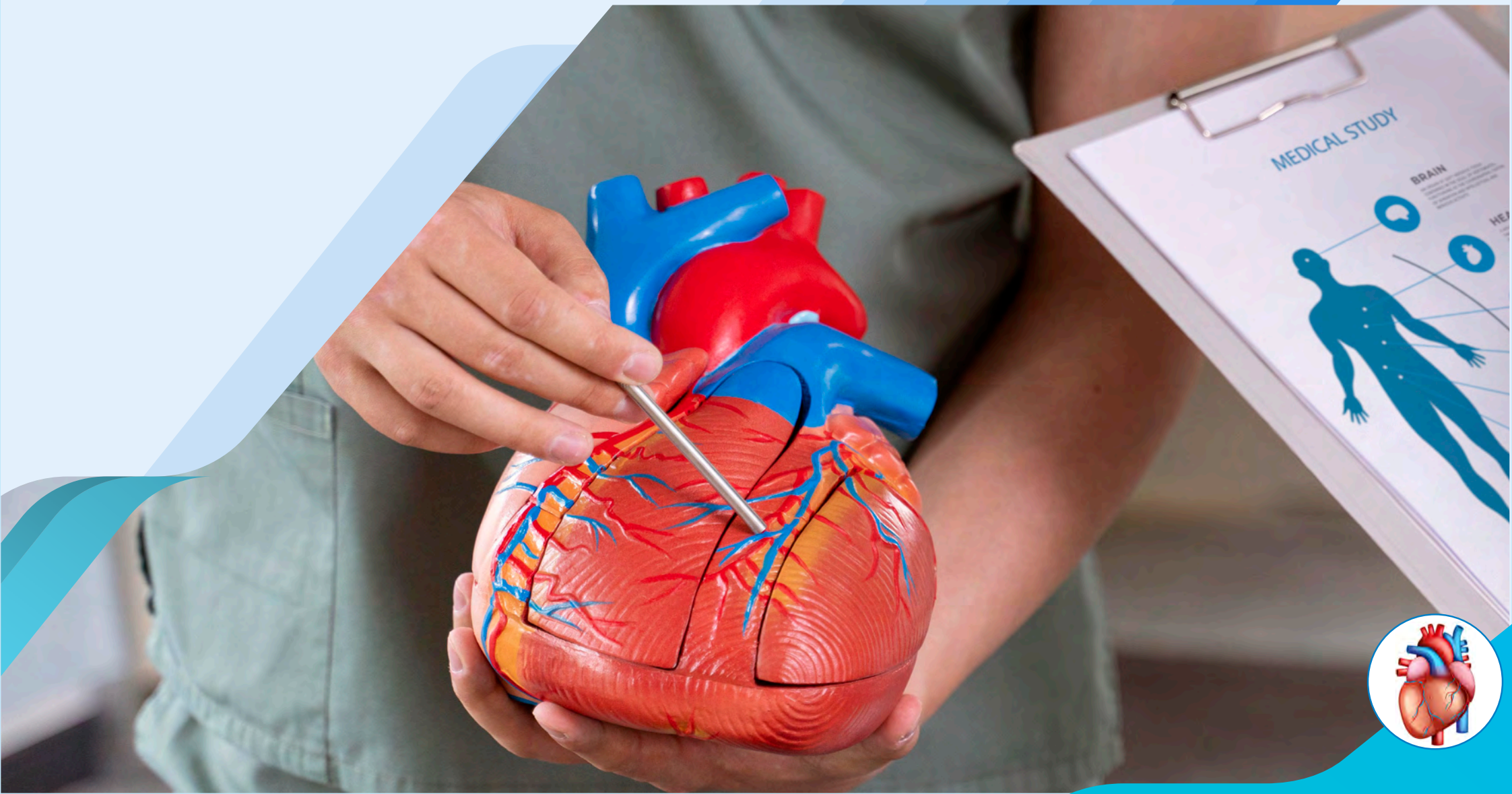
MICRO LABS

CARSYON 1

Algorithms in CARDIOLOGY



Preface

Cardiovascular disease (CVD) remains one of the leading causes of mortality and morbidity worldwide. The effective management of CVD hinges on the timely application of evidence-based practices and structured clinical decision-making.

This compilation of cardiology algorithms has been carefully developed to provide clarity, simplicity, and clinical utility. These algorithms are not meant to replace medical expertise, but rather to support clinical judgment by offering a practical framework for decision-making.

In today's rapidly evolving medical landscape, staying abreast of numerous innovations and therapeutic advancements is a constant challenge. These algorithms are designed to serve as reliable tools, helping healthcare professionals deliver high-quality cardiovascular care with confidence and consistency.

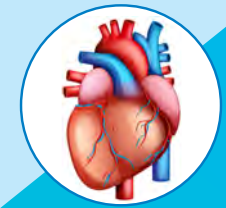


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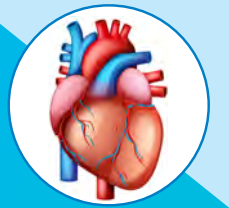
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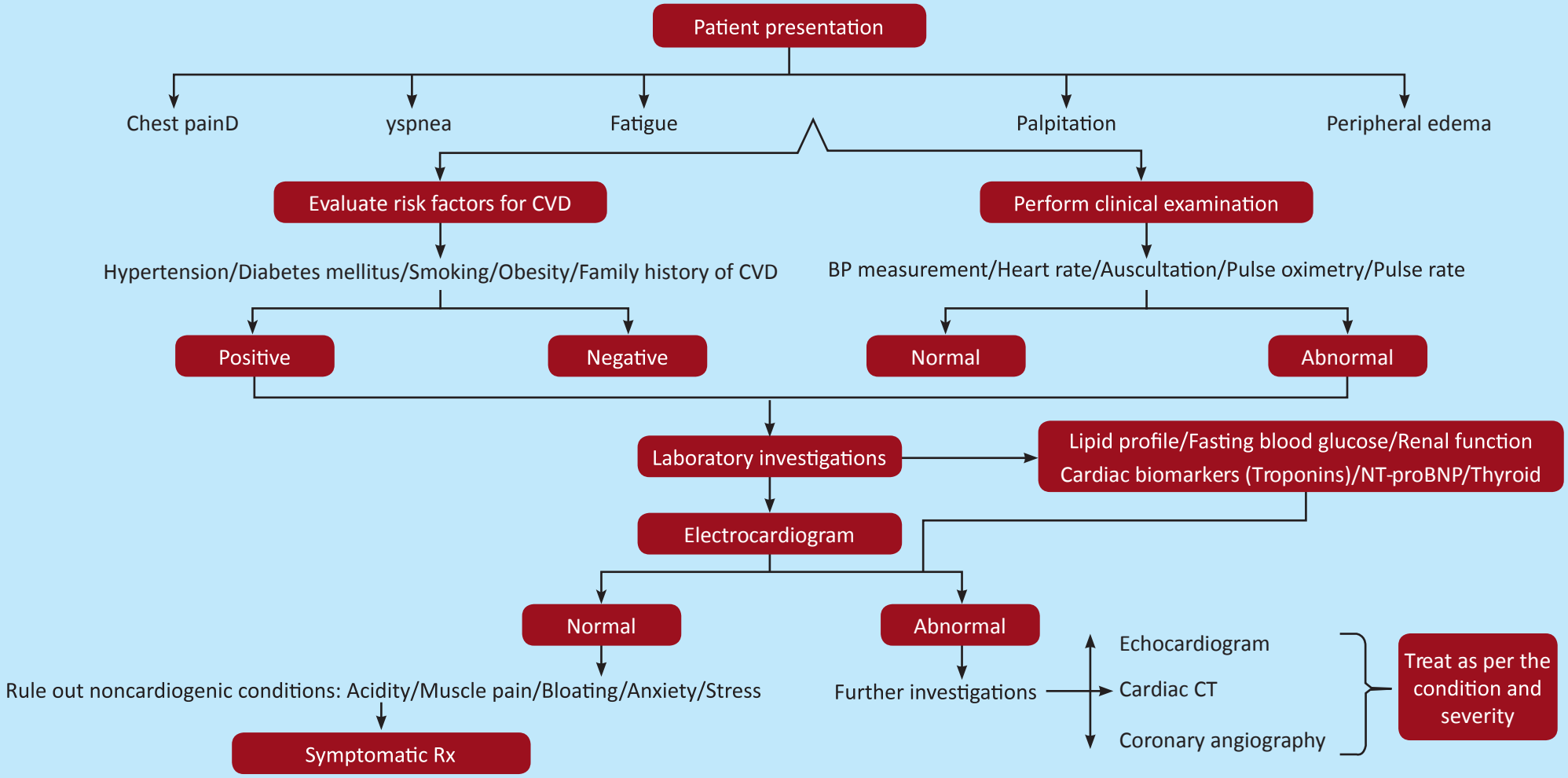
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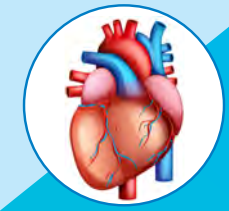
Algorithm for Initial Assessment



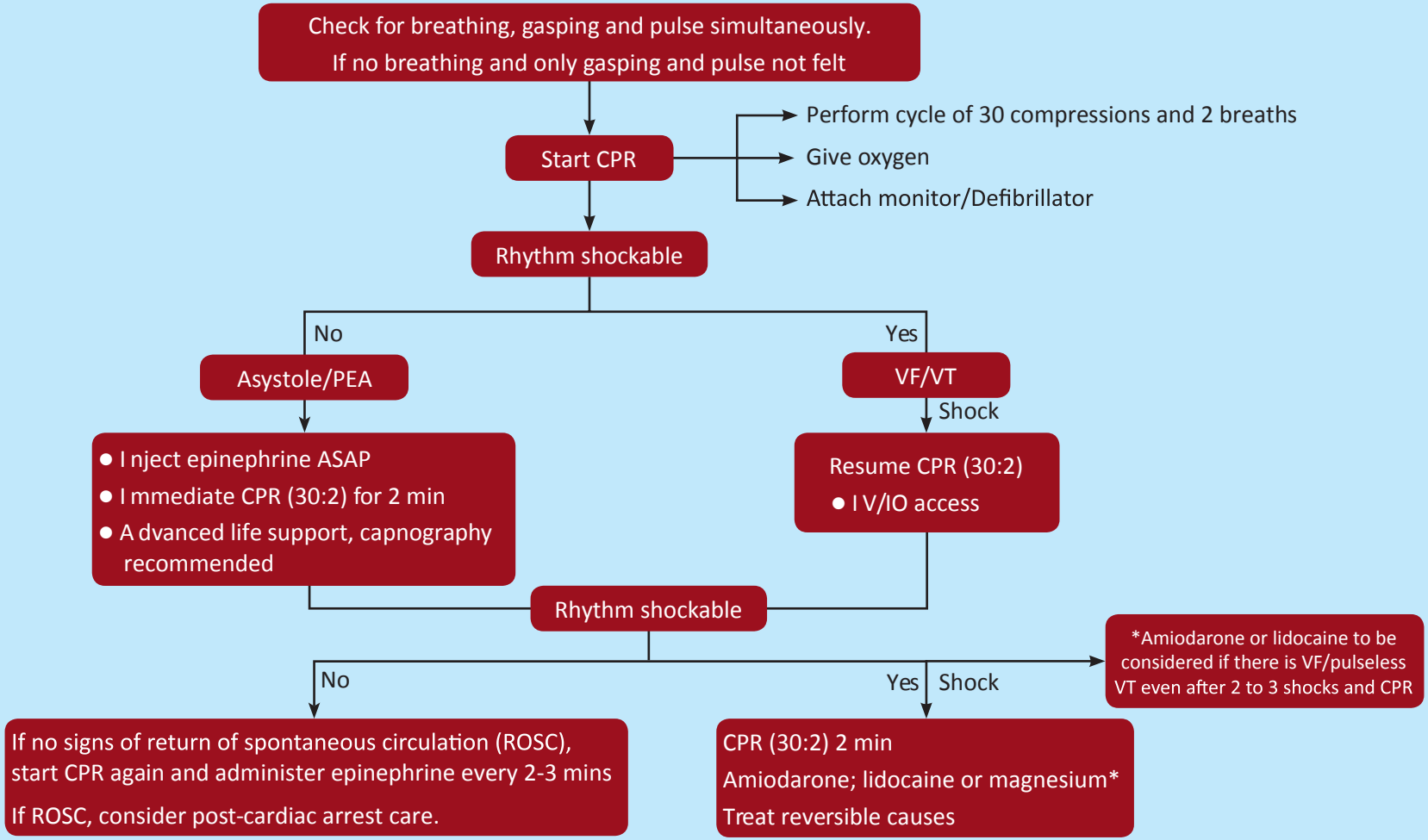
CVD: Cardiovascular disease; BP: Blood pressure; NT-proBNP: N-terminal pro-B-type natriuretic peptide; CT: Computed tomography.

Sources

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Cardiac Arrest

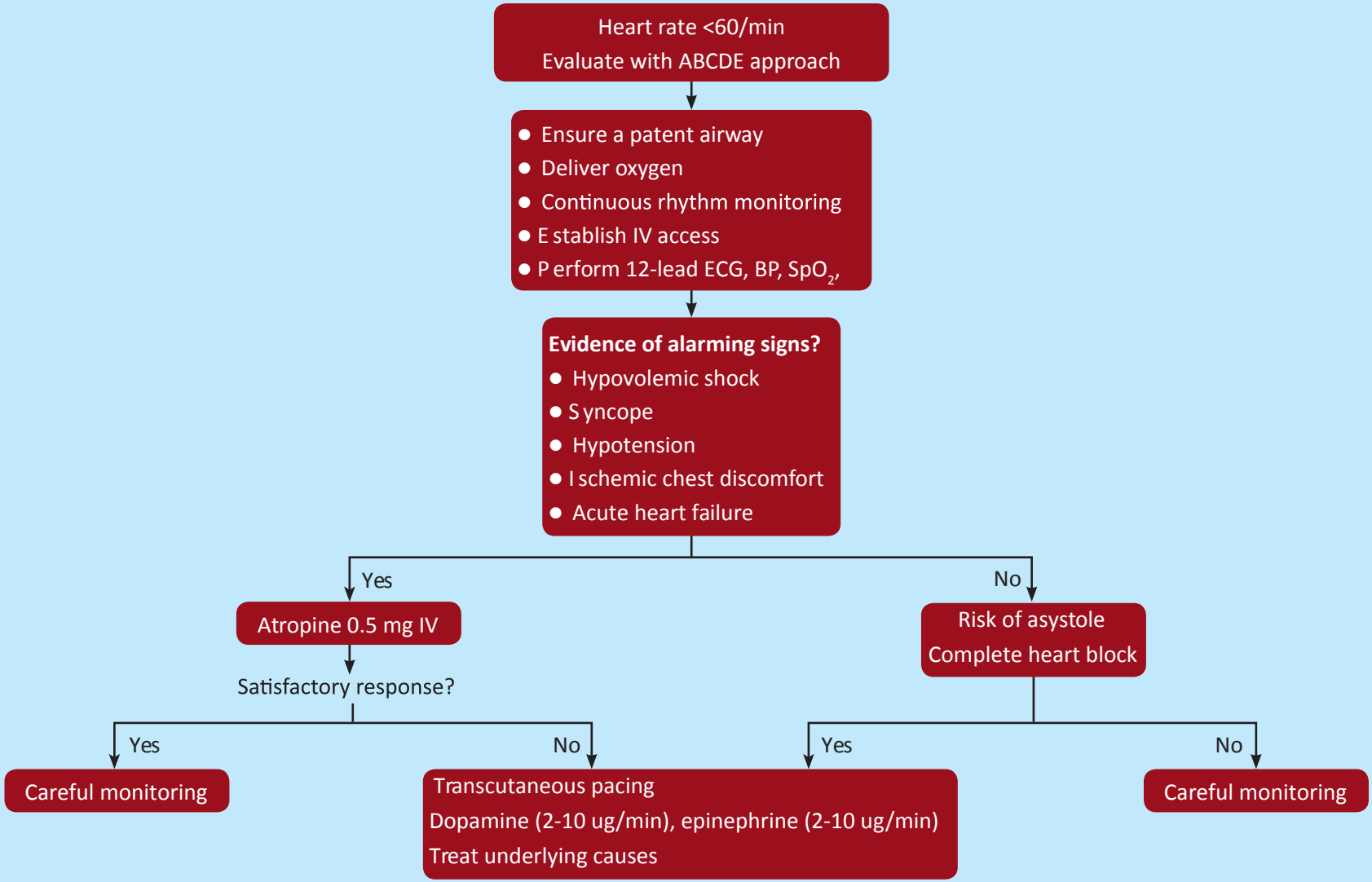


CPR: Cardiopulmonary resuscitation; PEA: Pulseless electrical activity; VF: Ventricular fibrillation; VT: Ventricular tachycardia; ASAP: As soon as possible; IV: Intravenous; IO: Intraosseus.

Sources

1. Pareek N, Kordis P, Webb I, et al. Contemporary management of out-of-hospital cardiac arrest in the Cardiac Catheterisation Laboratory: current status and future directions. Interv Cardiol. 2019;14(3):113-23.
2. G oyal A, Sciammarella JC, Cusick AS, Patel PH. Cardiopulmonary resuscitation. [Updated 2023 May 29]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK470402/
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Adult Bradycardia

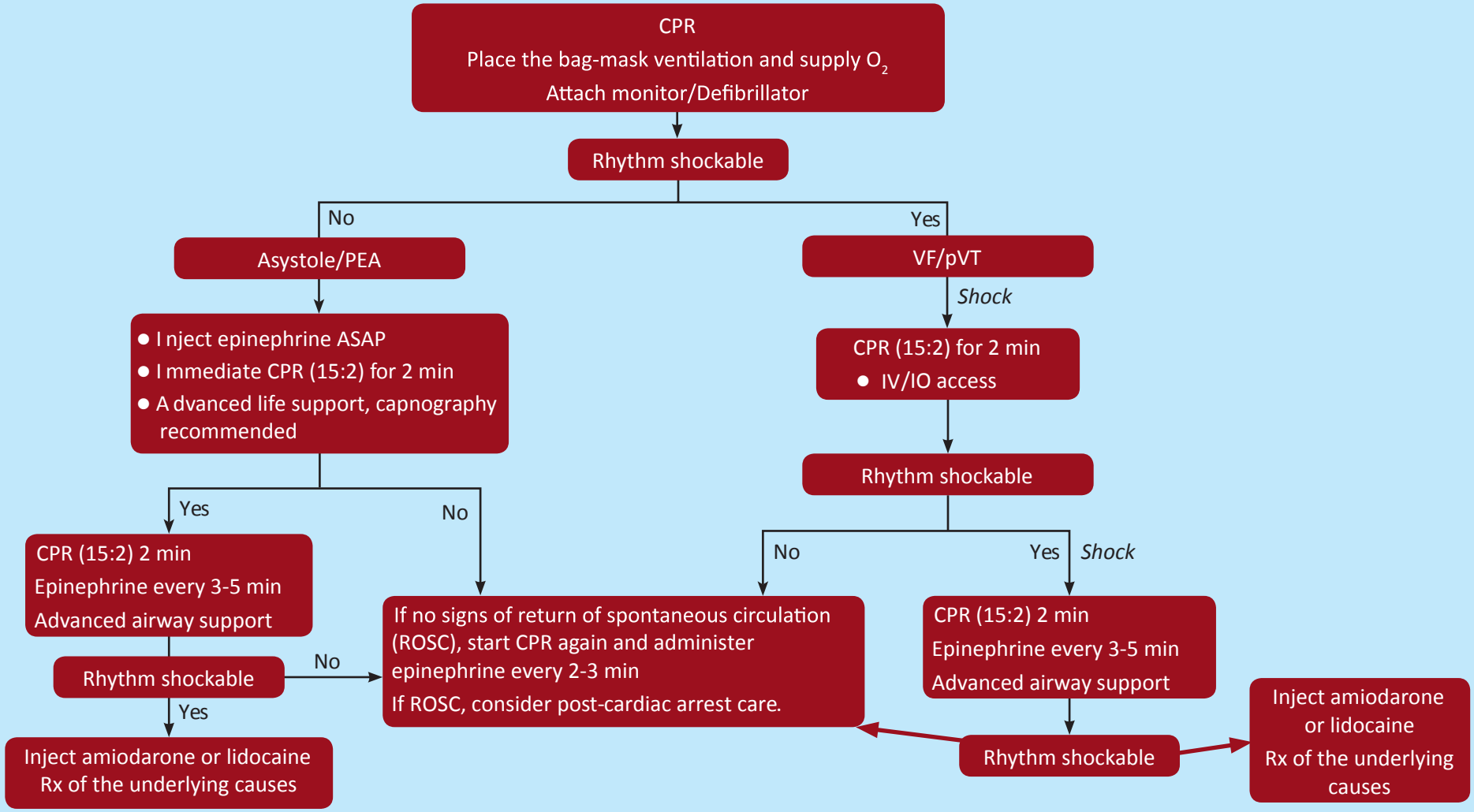


ABCDE: Airway, Breathing, Circulation, Disability, Exposure; IV: Intravenous; ECG: Electrocardiogram; BP: Blood pressure; SpO2: Oxygen saturation.

Sources

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Pediatric Cardiac Arrest

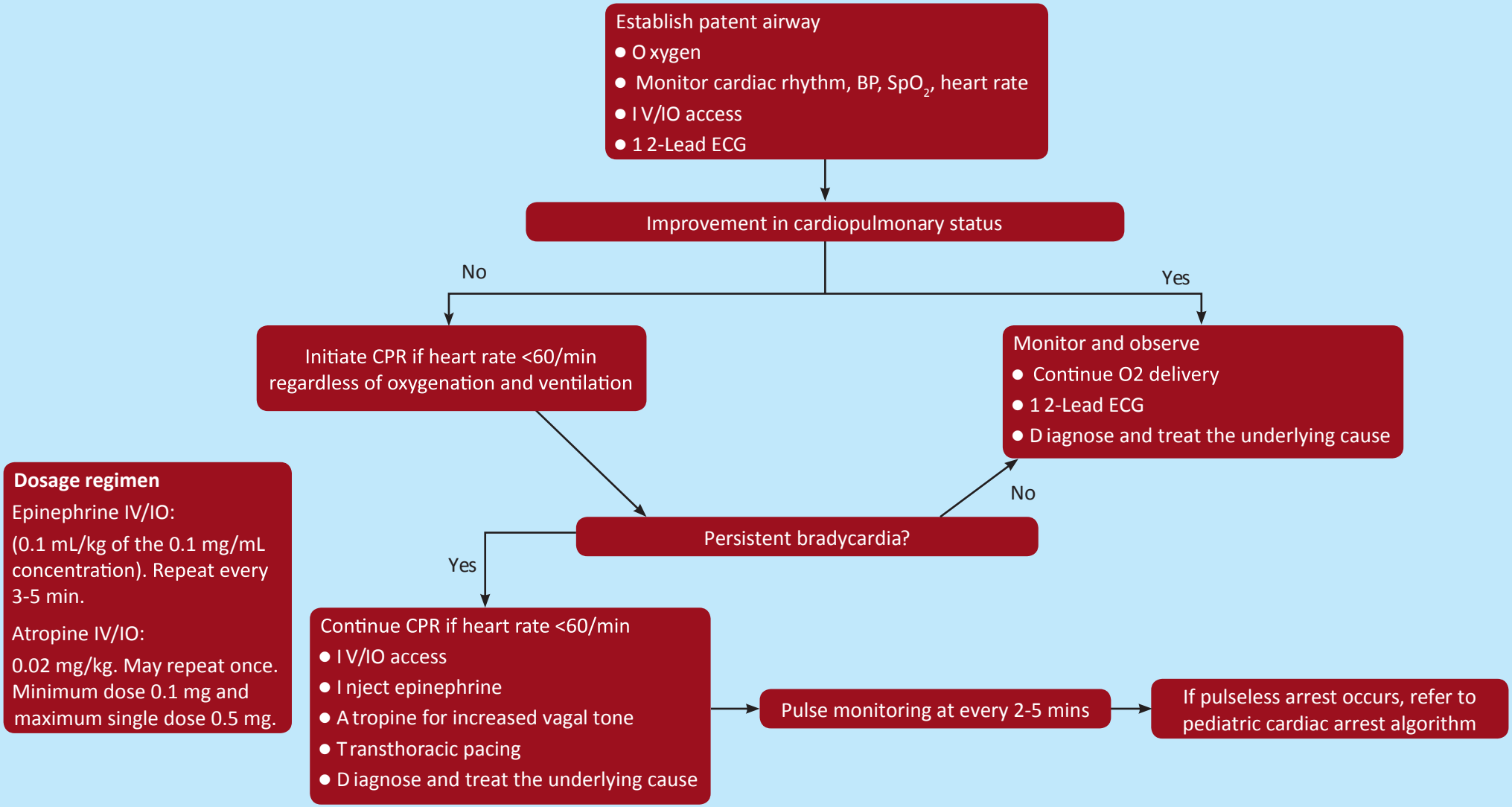


CPR: Cardiopulmonary resuscitation; O2: Oxygen; PEA: Pulseless electrical activity; ASAP: As soon as possible; VF: Ventricular fibrillation; pVT: Pulseless ventricular tachycardia; IV: Intravascular; IO: Intraosseous.

Sources

1. M organ RW, Kirschen MP, Kilbaugh TJ, Sutton RM, Topjian AA. Pediatric in-hospital cardiac arrest and cardiopulmonary resuscitation in the United States: A review. JAMA Pediatr. 2021;175(3):293-302.
2. G oyal A, Sciammarella JC, Cusick AS, Patel PH. Cardiopulmonary resuscitation. [Updated 2023 May 29]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK470402/
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Pediatric Bradycardia with a Pulse and Hypoperfusion

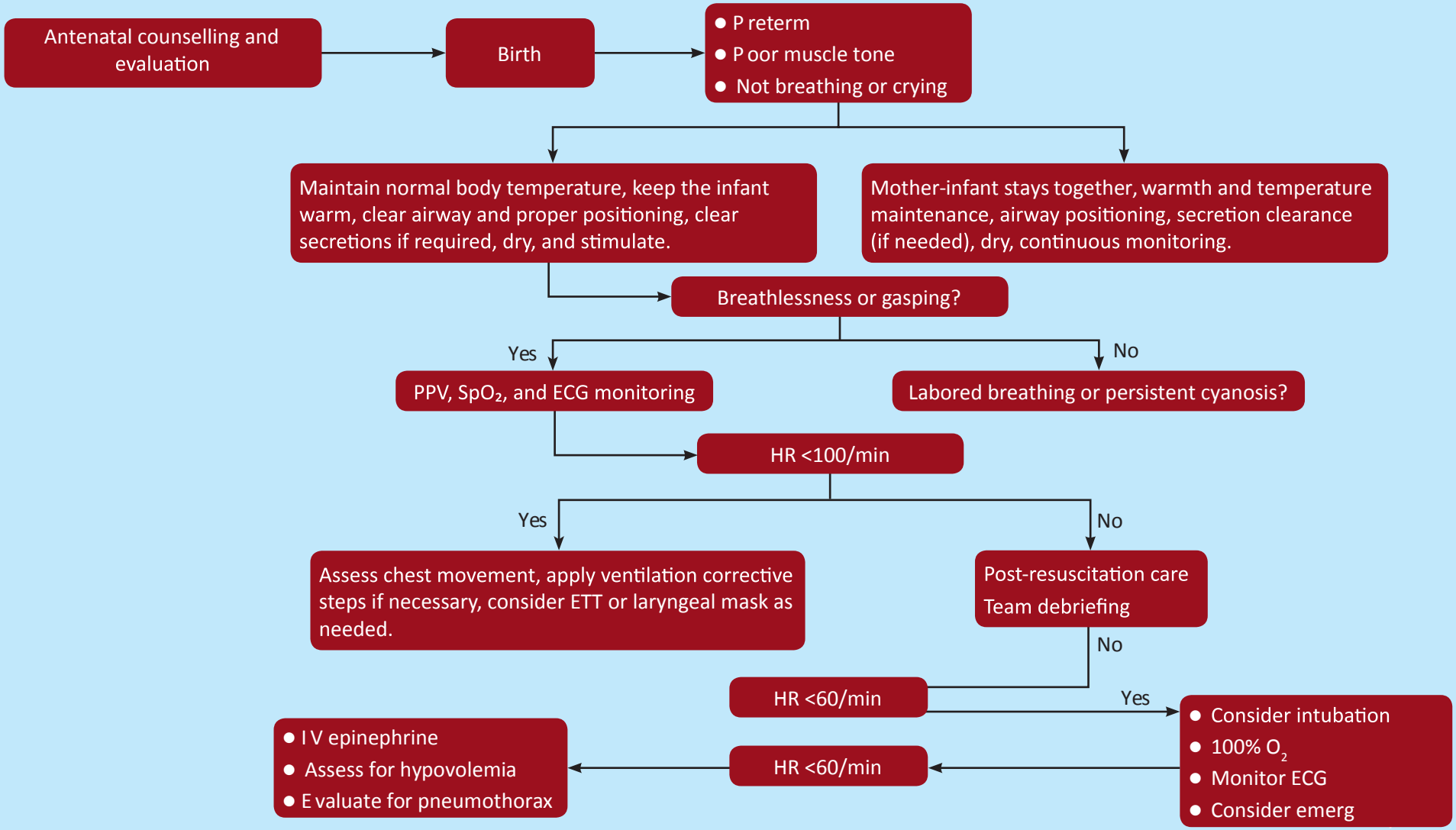


BP: Blood pressure; SpO2: Oxygen saturation; IV: Intravascular; IO: Intraosseous; CPR: Cardiopulmonary resuscitation.

Sources

1. Topjian AA, Raymond TT, Atkins D, et al; Pediatric Basic and Advanced Life Support Collaborators. Part 4: Pediatric basic and advanced life support: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation. 2020;142(16 Suppl 2):S469-S523.
2. Baruteau AE, Perry JC, Sanatani S, Horie M, Dubin AM. Evaluation and management of bradycardia in neonates and children. Eur J Pediatr. 2016;175(2):151-61.

Neonatal Resuscitation Algorithm

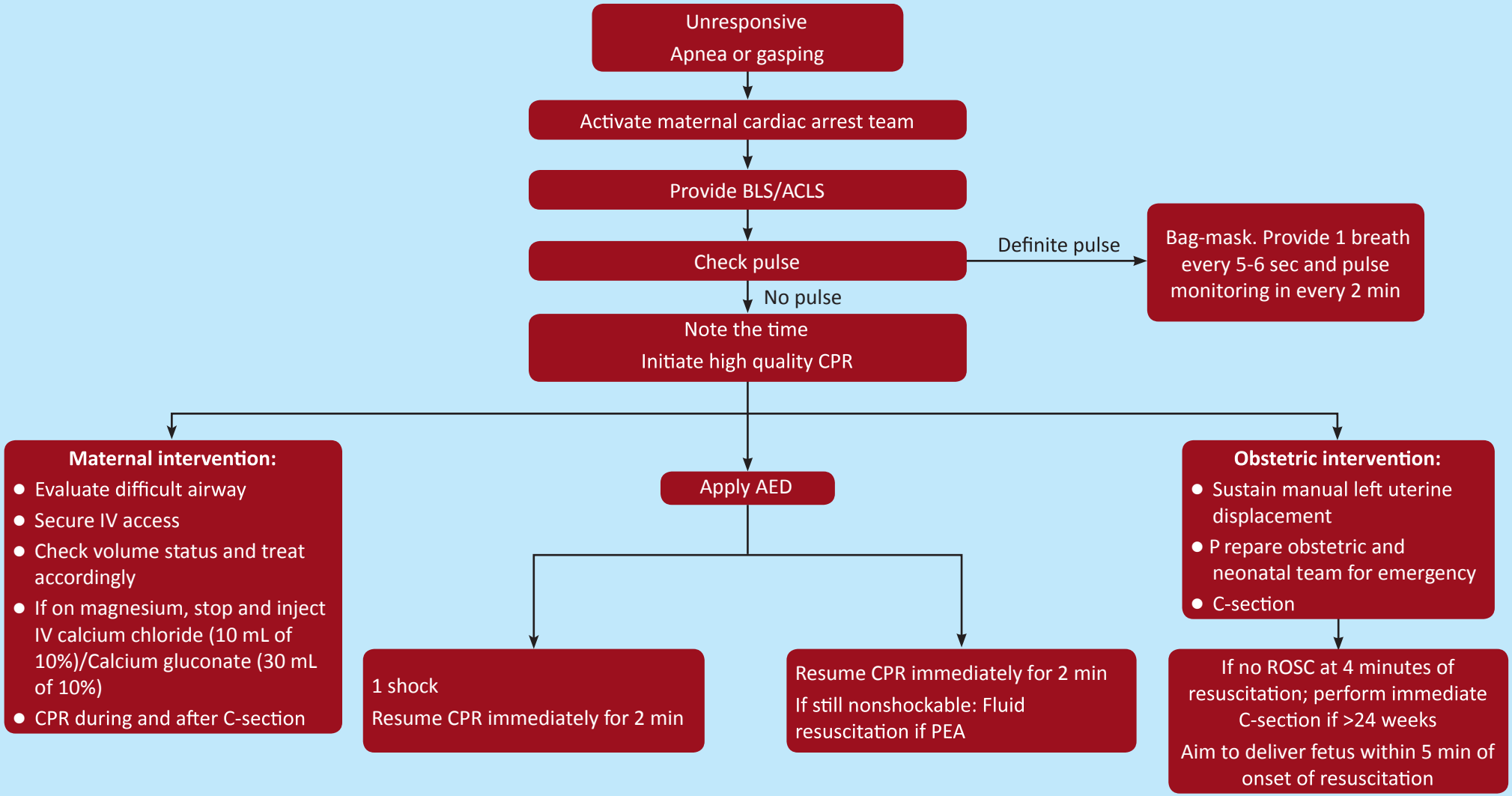


PPV: Positive-pressure ventilation; SpO_2: Oxygen saturation; ECG: Electrocardiogram; HR: Heart rate; ETT: Endotracheal tube; O_2: Oxygen; UVC: Umbilical venous catheter.

Sources

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2. Y amada NK, Szyld E, Strand ML, et al. 2023 American Heart Association and American Academy of Pediatrics focused update on neonatal resuscitation: an update to the American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. Circulation. 2024;149(1):e157-e166.
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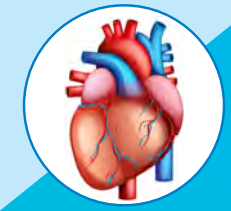
Cardiac Arrest in Pregnancy In-Hospital ACLS Algorithm



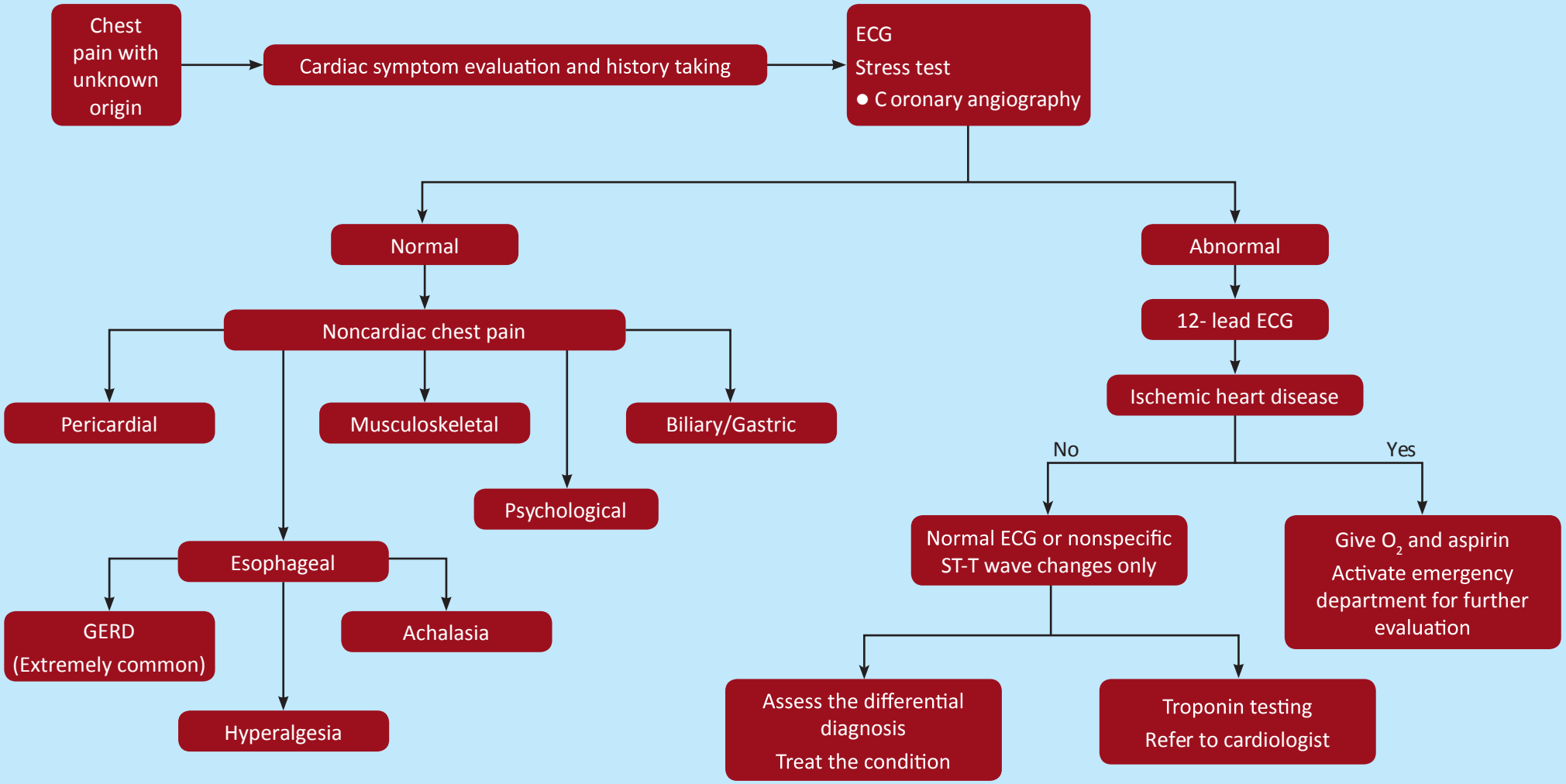
ACLS: Advanced cardiovascular life support; BLS: Basic life support, CPR: Cardiopulmonary resuscitation; IV: Intravenous; AED: Automated external defibrillator; PEA: Pulseless electrical activity; ROSC: Return of spontaneous circulation.

Sources

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Algorithm to Differentiate Between Cardiac and Noncardiac Causes of Chest Pain

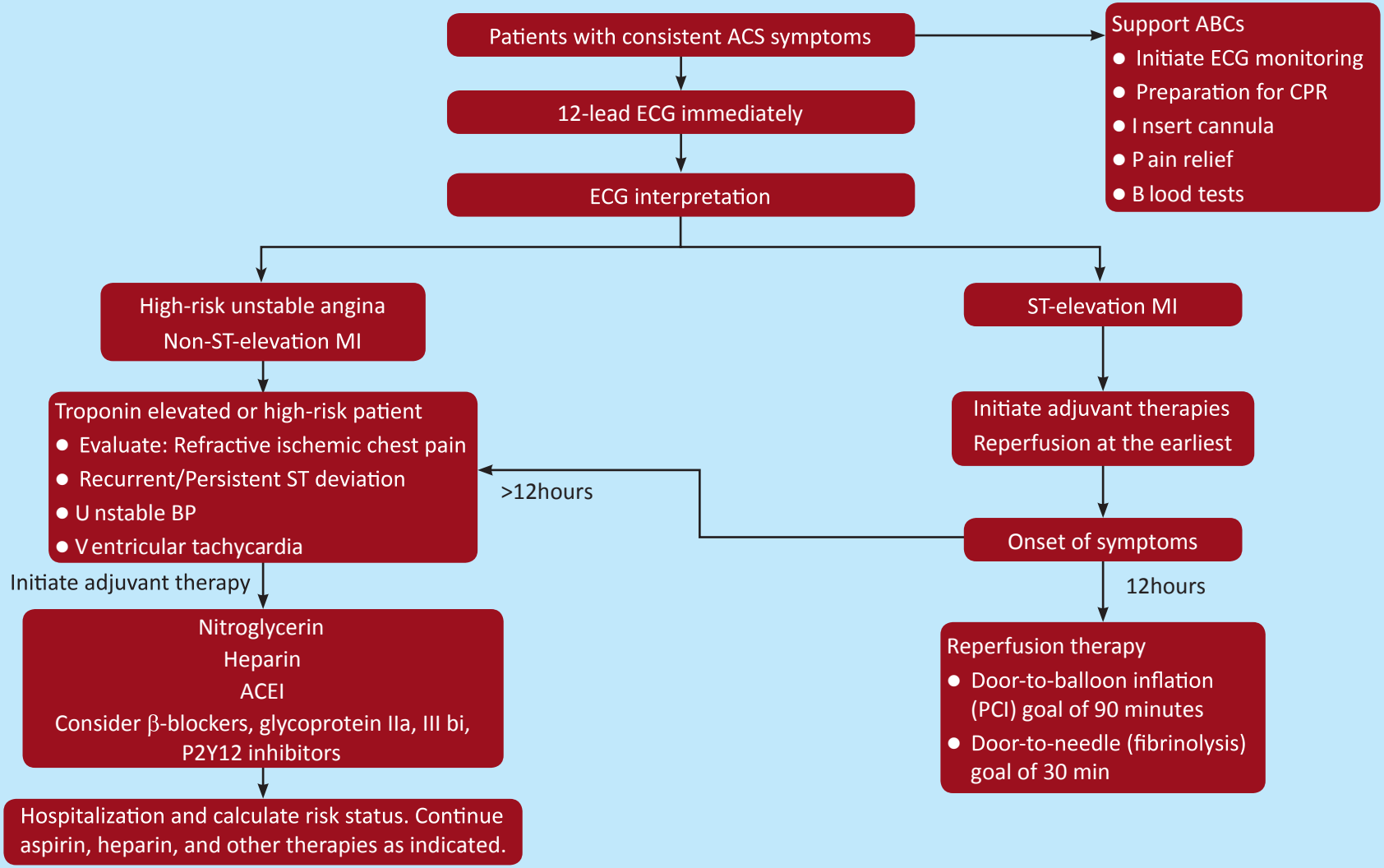


ECG: Electrocardiogram; GERD: Gastroesophageal reflux disease.

Sources

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Acute Coronary Syndrome Management Algorithm

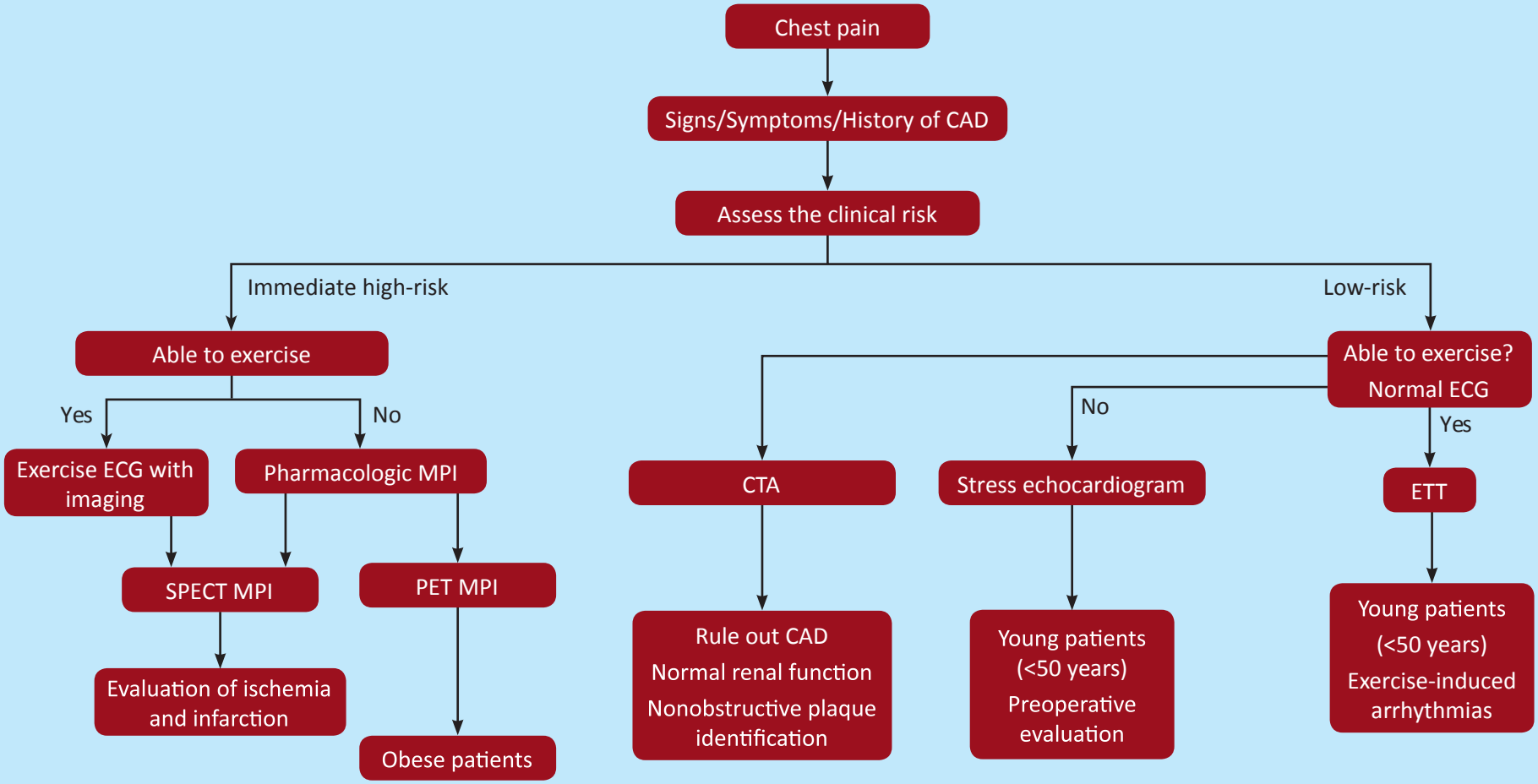


ACS: Acute coronary syndrome; ABCs: Airway, Breathing, and Circulation; ECG: Electrocardiogram; CPR: Cardiopulmonary resuscitation; MI: Myocardial infarction; BP: Blood pressure; ACEI: Angiotensin-converting enzyme inhibitor; PCI: Percutaneous coronary intervention.

Sources

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Diagnostic Tests: An Algorithm to Guide the Selection of Appropriate Cardiac Imaging

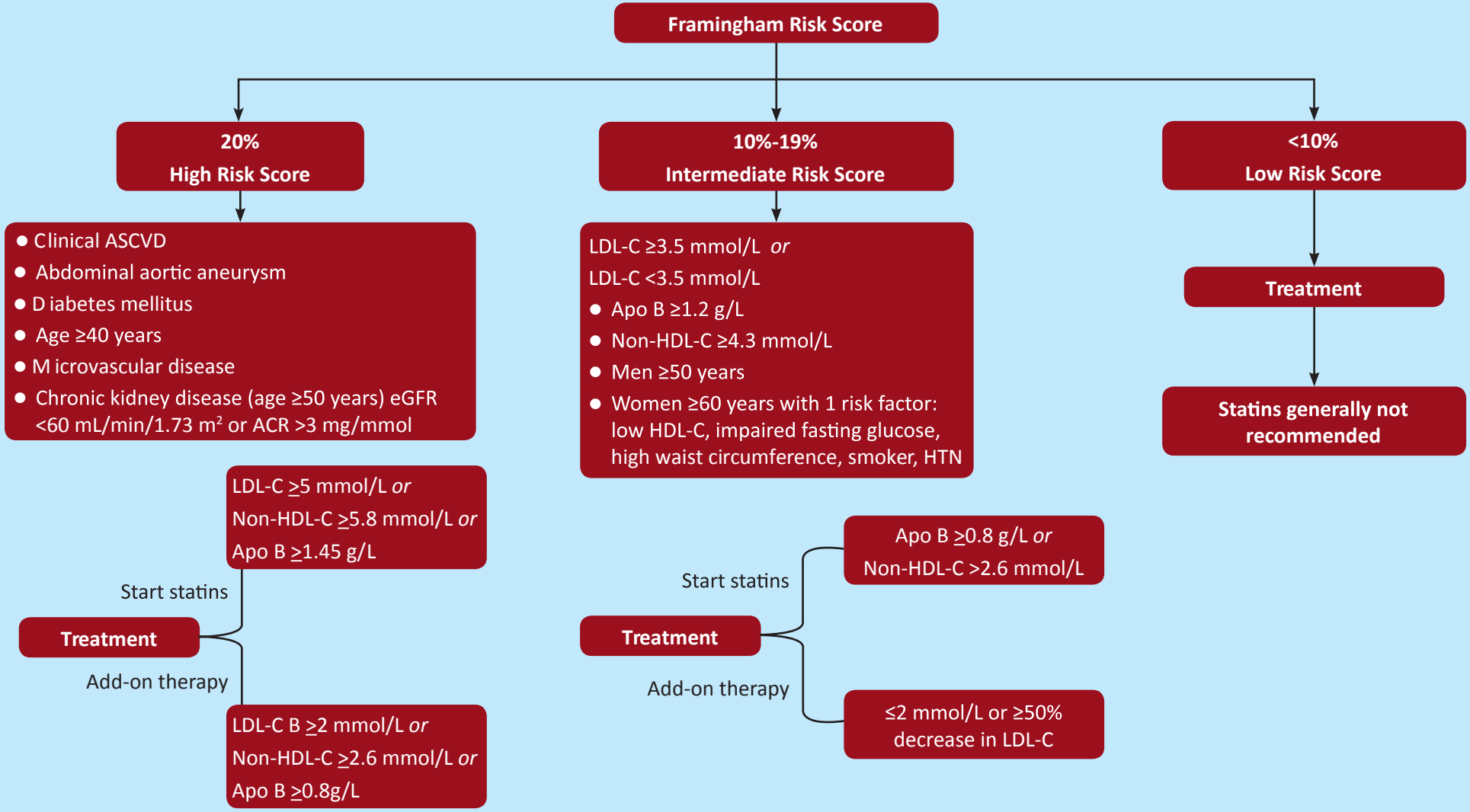


CAD: Coronary artery disease; ECG: Electrocardiogram; MPI: Myocardial perfusion imaging; SPECT: Single-photon emission computed tomography; PET: Positron emission tomography; CTA: Computed tomography angiography; ETT: Exercise tolerance testing.

Sources

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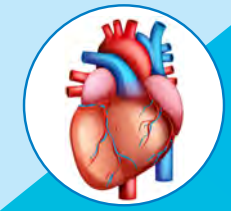
Cardiovascular Risk Assessment Scores

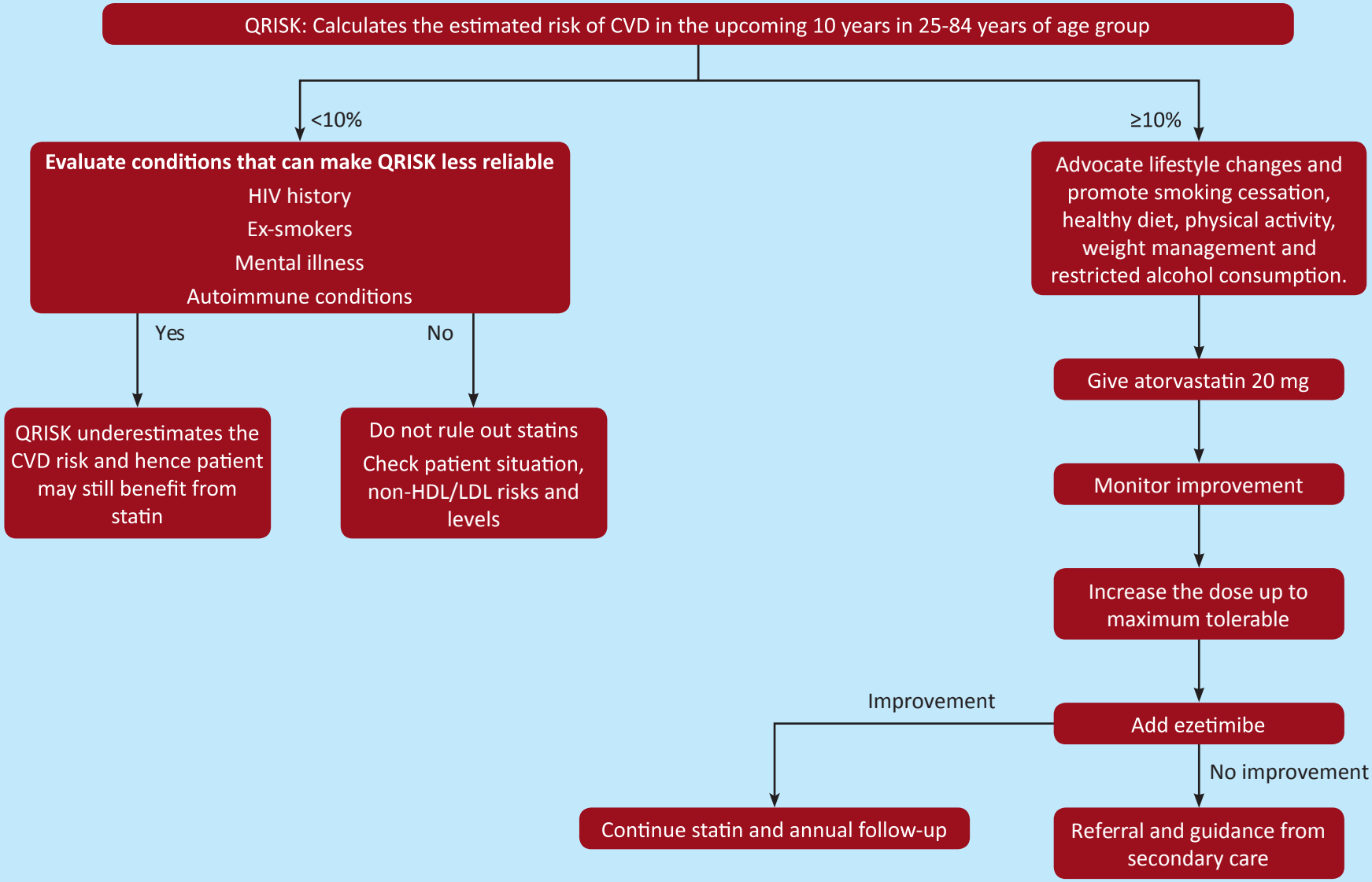


ASCVD: Atherosclerotic cardiovascular disease; eGFR: Estimated glomerular filtration rate; LDL-C: Low-density lipoprotein cholesterol; HDL-C: High-density lipoprotein cholesterol; Apo B: Apolipoprotein B; HTN: Hypertension.

Sources

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QRISK: QRESEARCH cardiovascular risk algorithm; CVD: Cardiovascular disease; HIV: Human immunodeficiency virus; HDL: High-density lipoprotein; LDL: Low-density lipoprotein.

Sources

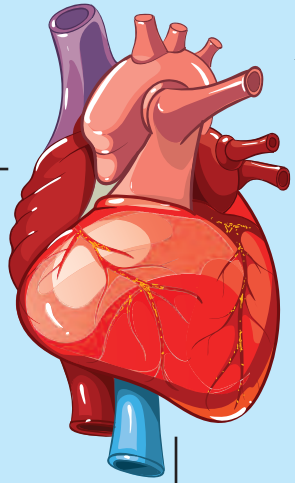
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Cardiac Biomarkers

HFrEF
BNP and NT-proBNP

Thrombosis-Fibrinolysis
- D -dimers
- F ibrinogen

Myocardial injury
- T roponin
- Creatine kinase
- H -type FABP



Inflammation
- hsCRP
- I L-6
- TNF-α
- P rocalcitonin

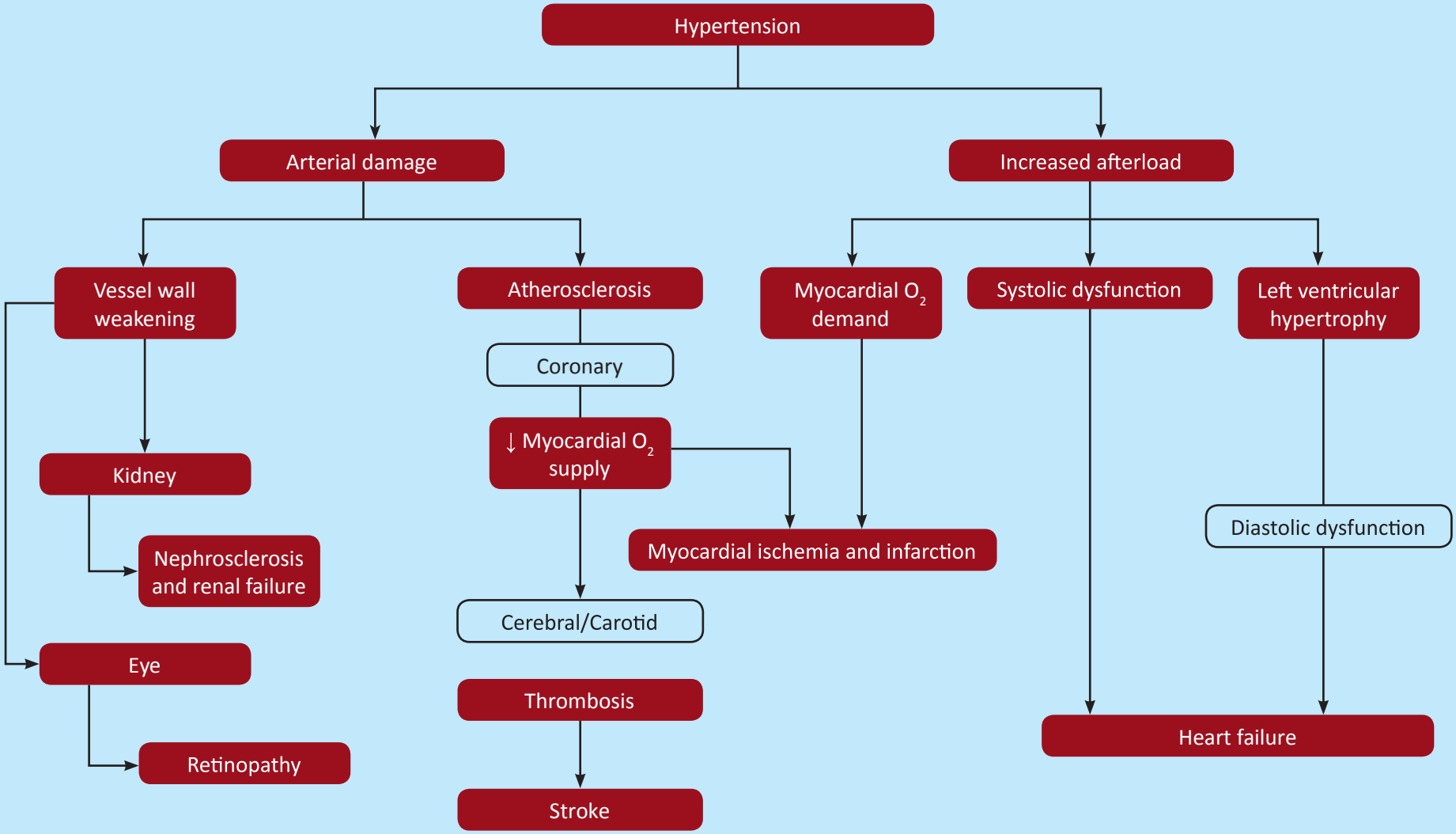
Oxidative stress
- Uric acid
- M yeloperoxidase

HFrEF: Heart failure with reduced ejection fraction; BNP: B-type natriuretic peptide; NT-proBNP: N-terminal proBNP; FABP: Fatty acid binding protein; hsCRP: High-sensitivity C-reactive protein; IL-6: Interleukin 6; TNF: Tumor necrosis factor.

Sources

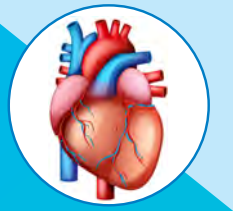
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Hypertension Fate

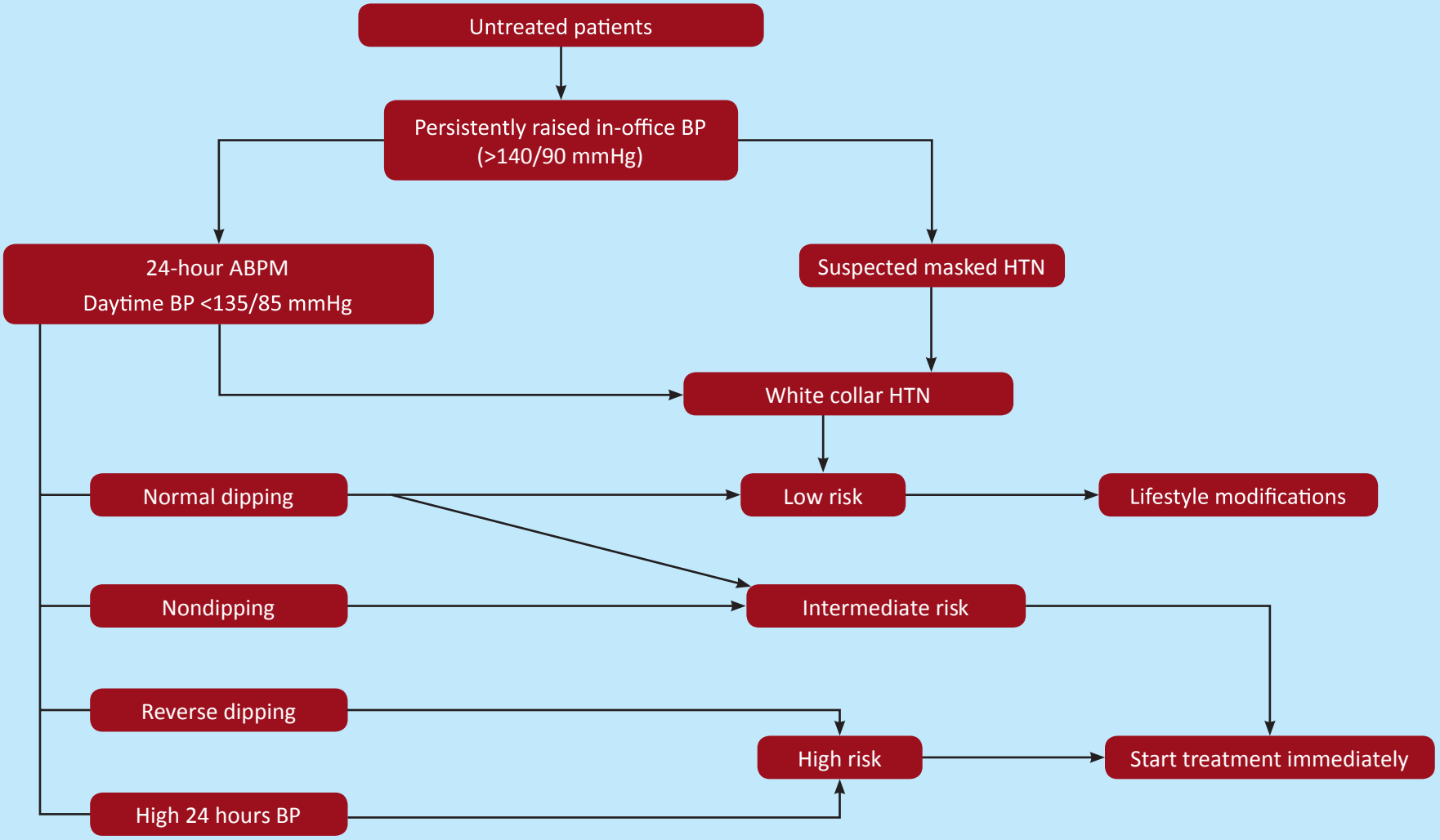


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White Collar Hypertension

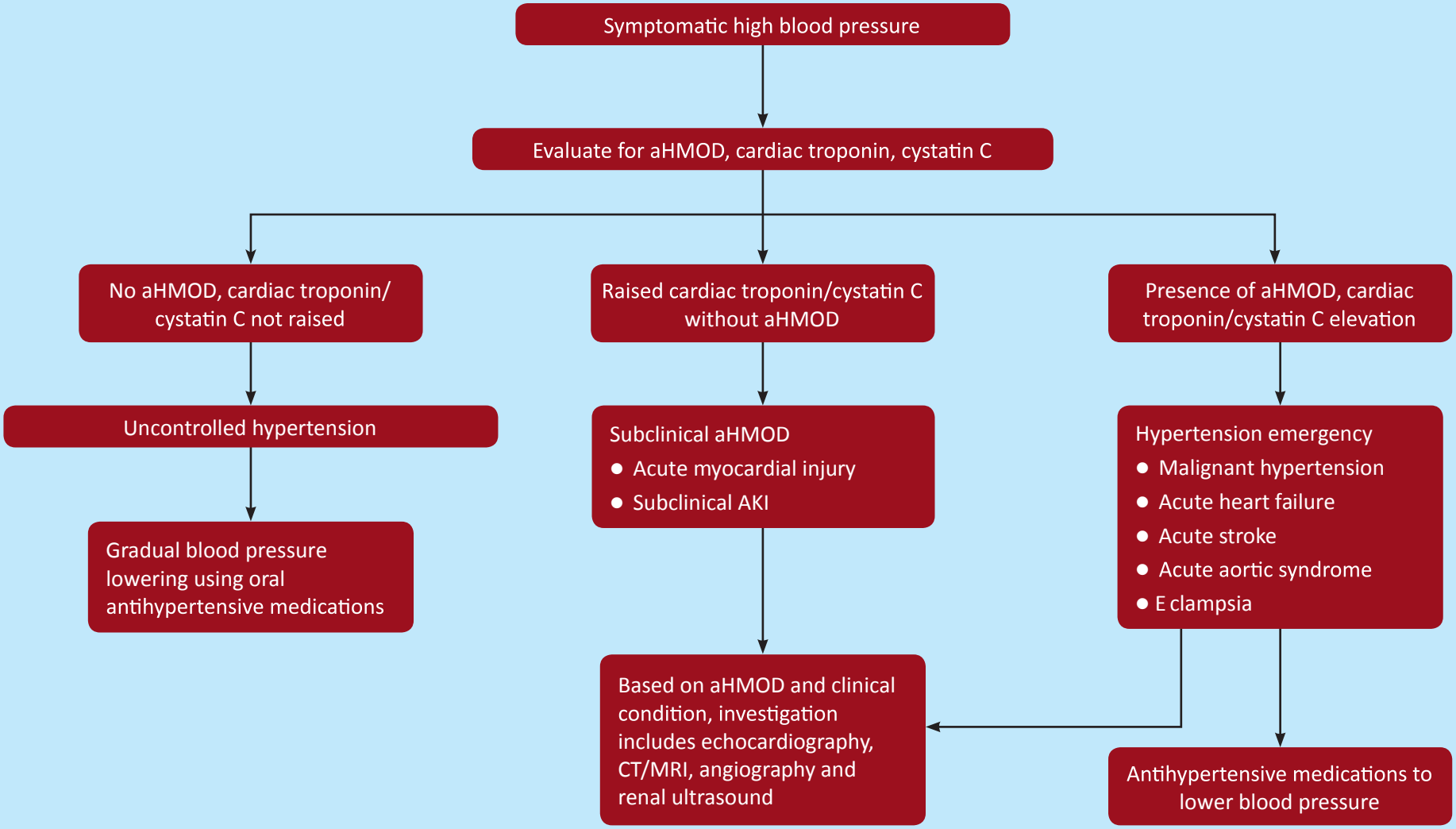


BP: Blood pressure; ABPM: Ambulatory blood pressure monitoring; HTN: Hypertension.

Sources

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Classification and Evaluation Algorithm

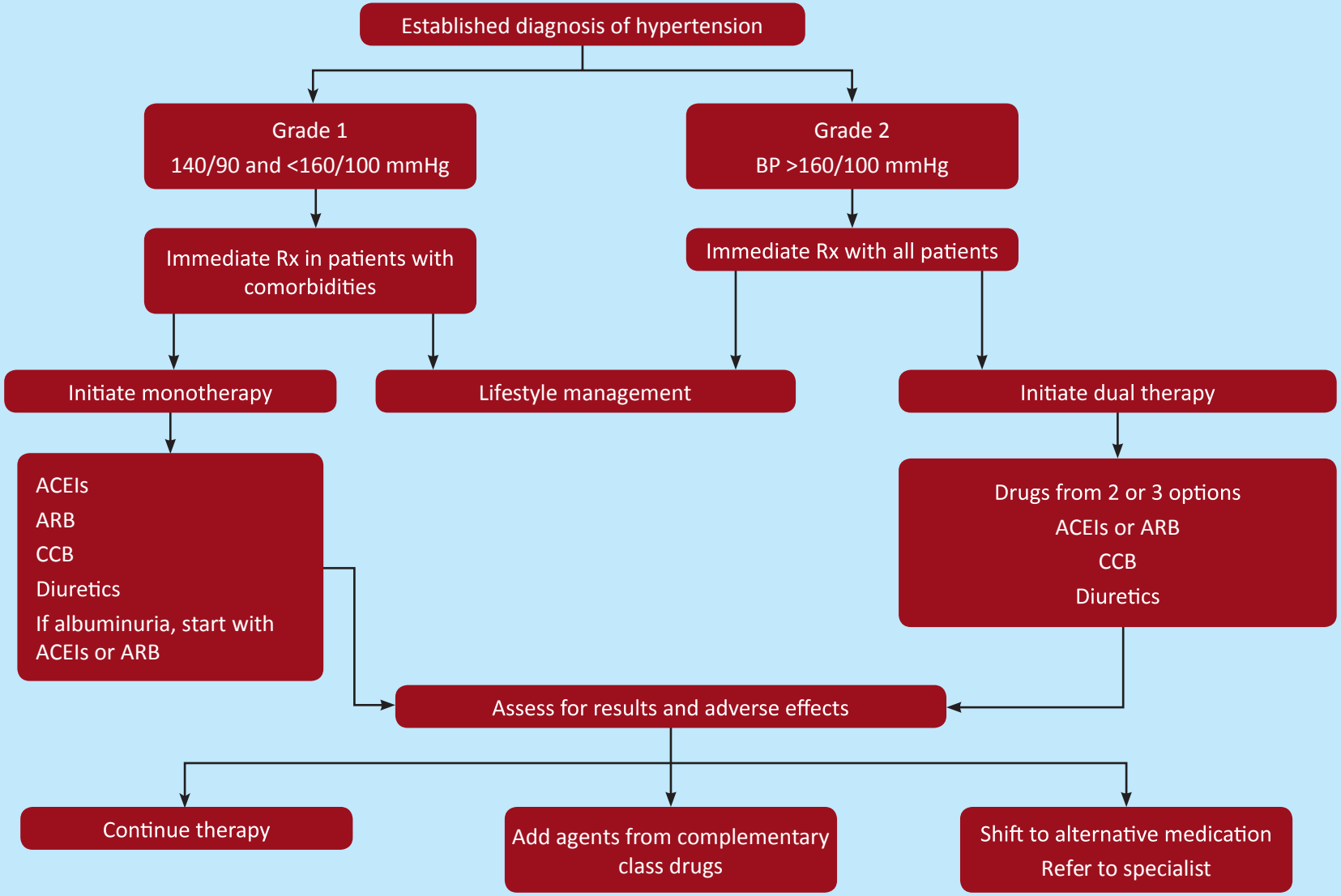


aHMOD: Acute hypertension-mediated organ damage; AKI: acute kidney injury; CT: Computed tomography; MRI: Magnetic resonance imaging.

Sources

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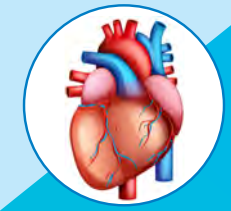
Hypertension Management

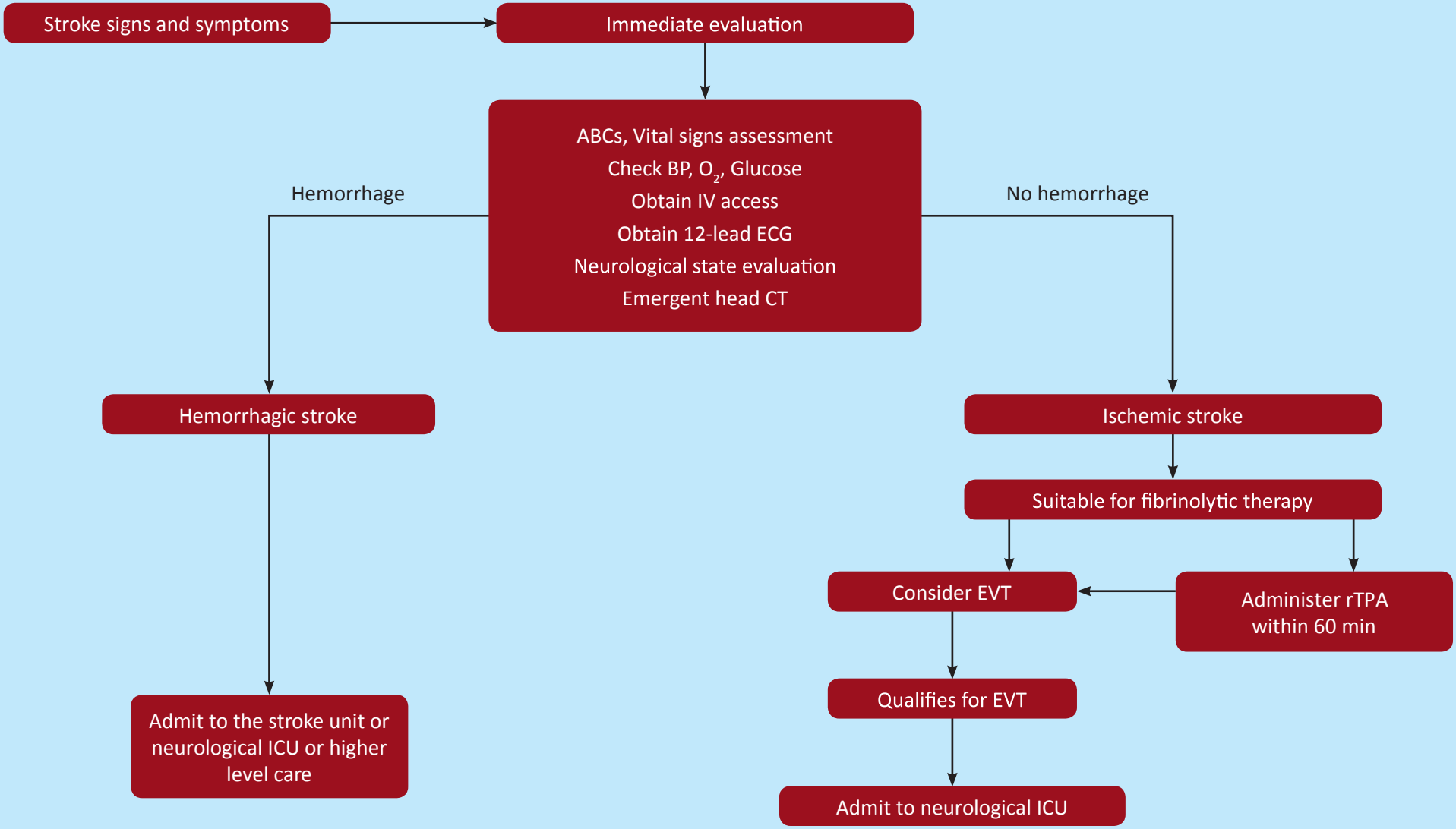


BP: Blood pressure; ACEIs: Angiotensin-converting enzyme inhibitors; ARB: Angiotensin II receptor blockers; CCBs: Calcium channel blockers.

Sources

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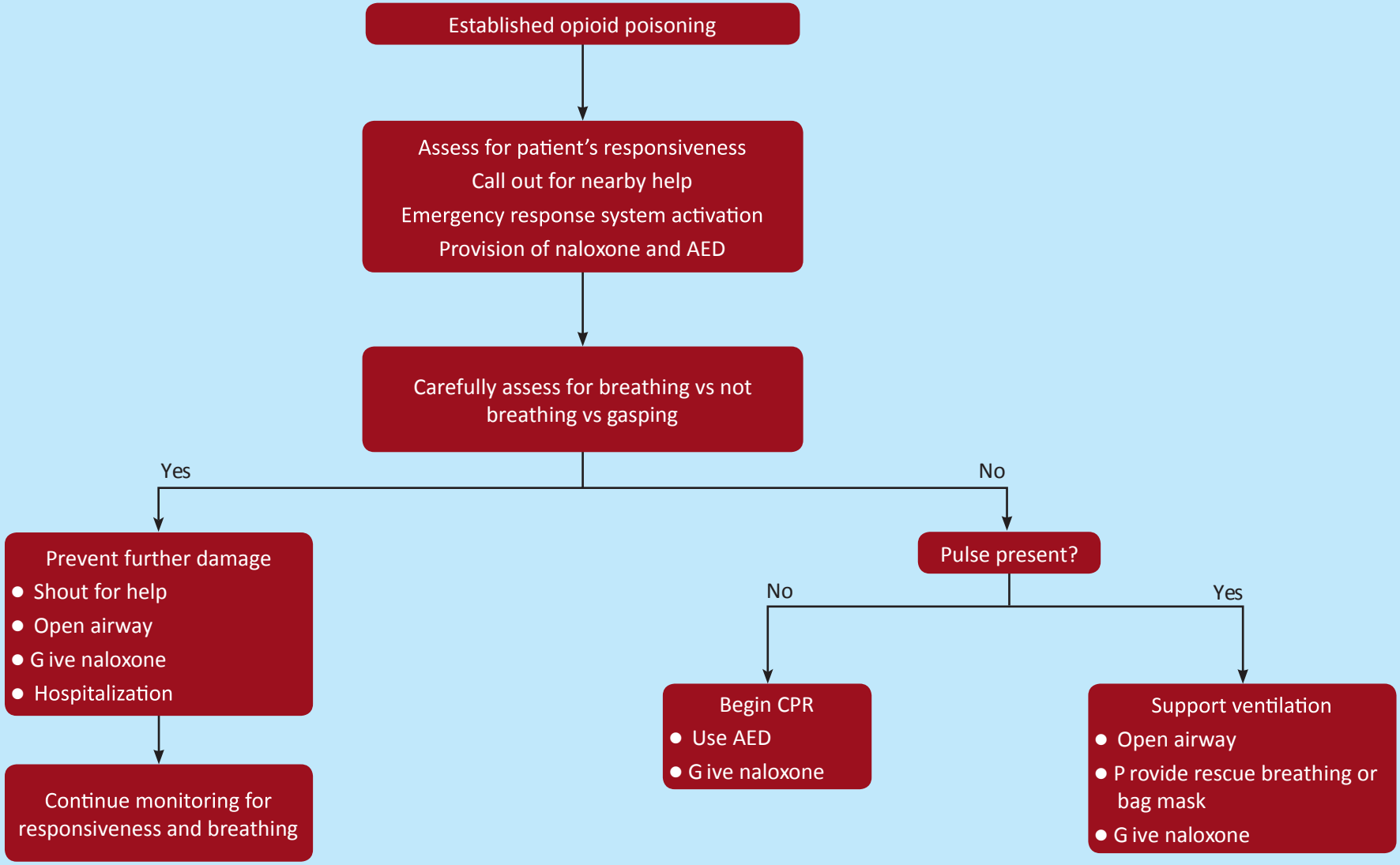


ABCs: Airway, Breathing, and Circulation; BP: Blood pressure; ECG: Electrocardiogram; CT: Computed tomography; ICU: Intensive care unit; EVT: Endovascular thrombectomy; rTPA: Recombinant tissue plasminogen activator.

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Opioid-Associated Out-of-Hospital Cardiac Arrest



AED: Automated external defibrillator; CPR: Cardiopulmonary resuscitation.

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